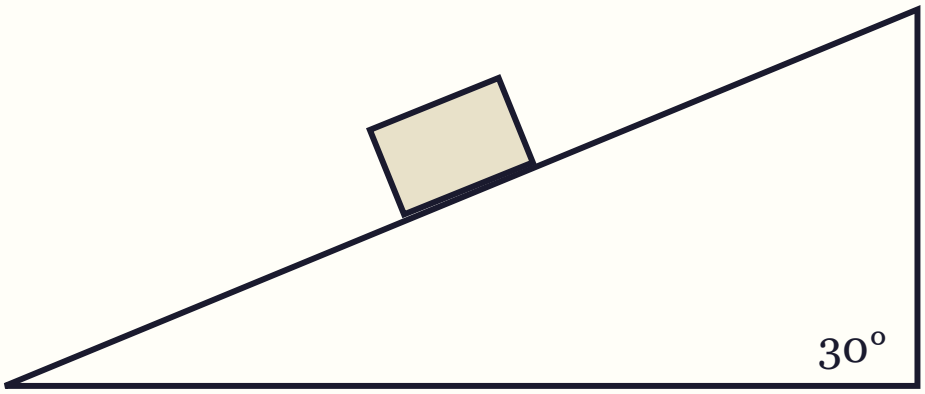
Kinematics

Mistake 1

Block on a rough incline

Coaching DPP · DPP 4 Q12 · Silly calculation

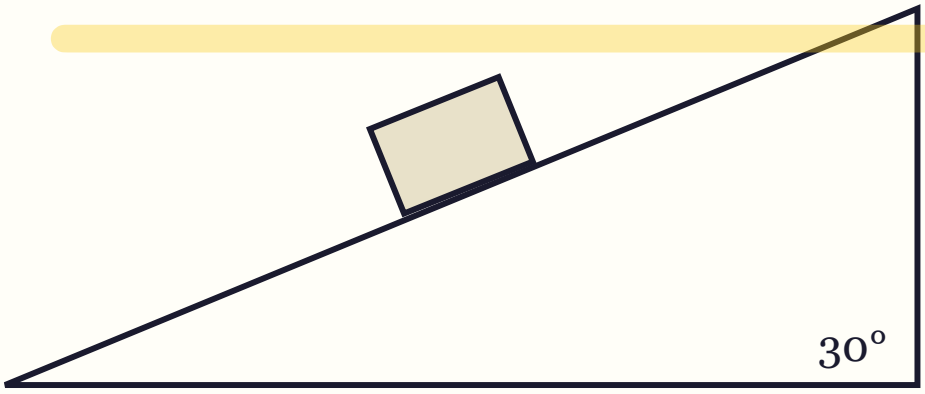
Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)



(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)



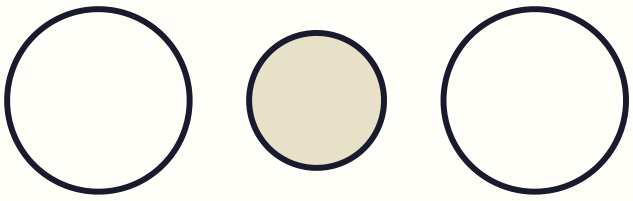
(a) 3.3 (b) 4.0
(c) 5.0 (d) 6.7

Mistake 2

Moles of oxygen in CO_2

Mock · AITS 3 · Concept gap

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



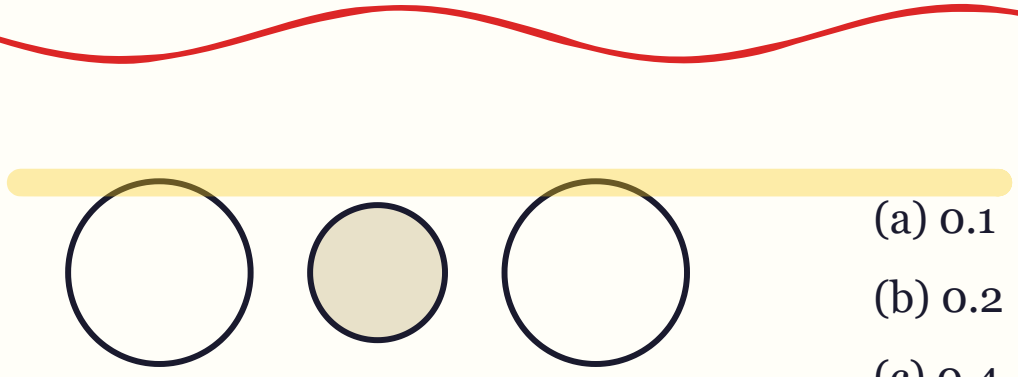
(a) 0.1

(b) 0.2

(c) 0.4

I took the weight as mg and forgot to resolve it along the incline.

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



(a) 0.1

(b) 0.2

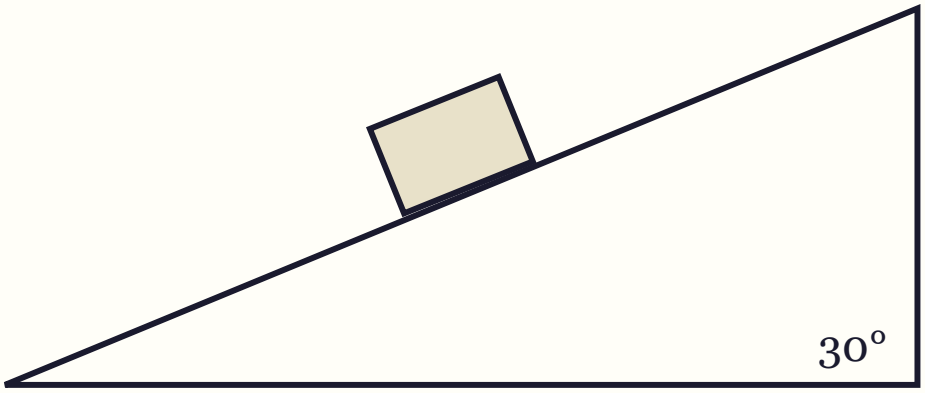
(c) 0.4

Mistake 3

Projectile on an inclined plane

Coaching DPP · DPP 4 Q12 · Misread

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)

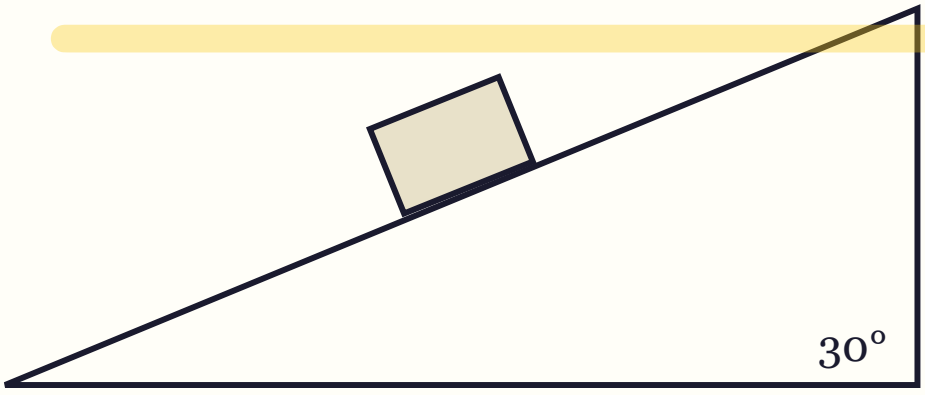


(a) 3.3 (b) 4.0

(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)



(a) 3.3 (b) 4.0

(c) 5.0 (d) 6.7

Mistake 4

Relative velocity of two trains

PYQ · JEE Main 2023 · Time pressure

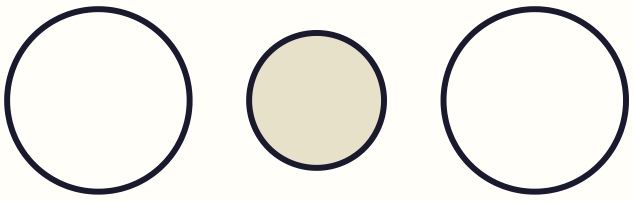
Ran out of time on the second part.

Mistake 5

Stopping distance with friction

Coaching DPP · DPP 4 Q12

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



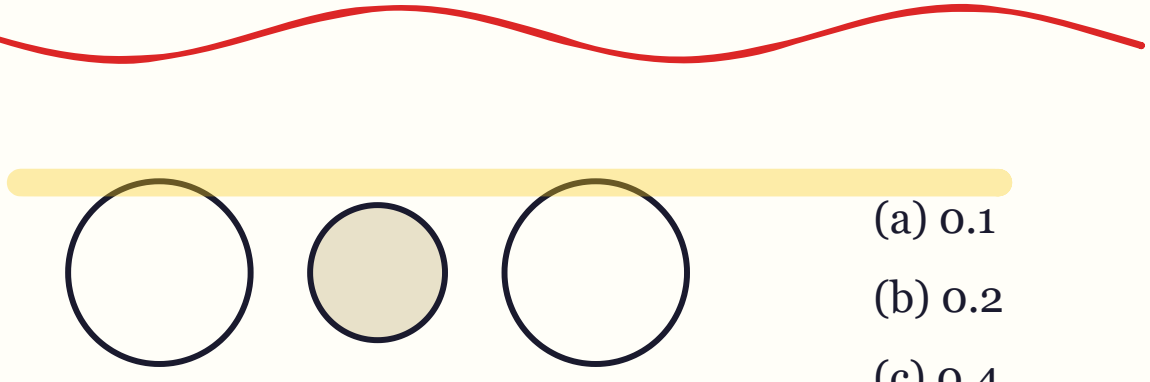
(a) 0.1

(b) 0.2

(c) 0.4

I took the weight as mg and forgot to resolve it along the incline.

Q7. How many moles of oxygen atoms are in 4.4 g of CO_2?



(a) 0.1

(b) 0.2

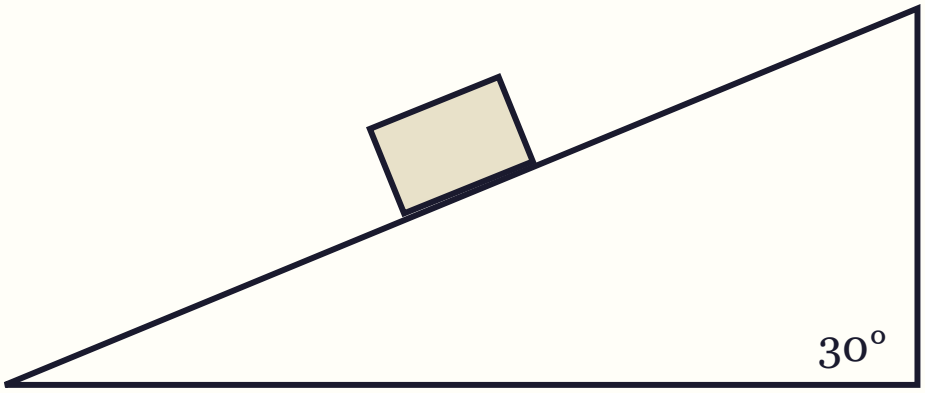
(c) 0.4

Mistake 6

Free fall from a moving balloon

Coaching DPP · DPP 4 Q12 · Forgot formula

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)

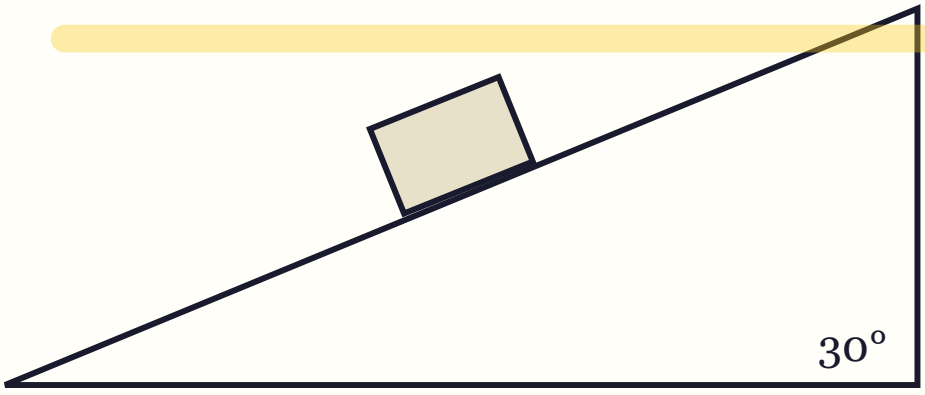


(a) 3.3 (b) 4.0

(c) 5.0 (d) 6.7

I took the weight as mg and forgot to resolve it along the incline.

Q12. A block of mass 2 kg slides down a rough incline of angle 30 degrees. The coefficient of friction is 0.2. Find its acceleration. ($g = 10\ m/s^2$)



(a) 3.3 (b) 4.0

(c) 5.0 (d) 6.7